4 (a) (i) A cubic container of sides s and cross-sectional area A contains a gas made up of a large number of molecules, each with mass m. One of these molecules has an initial velocity c and a velocity component u_x in the x direction, as shown in Fig. 4.1.

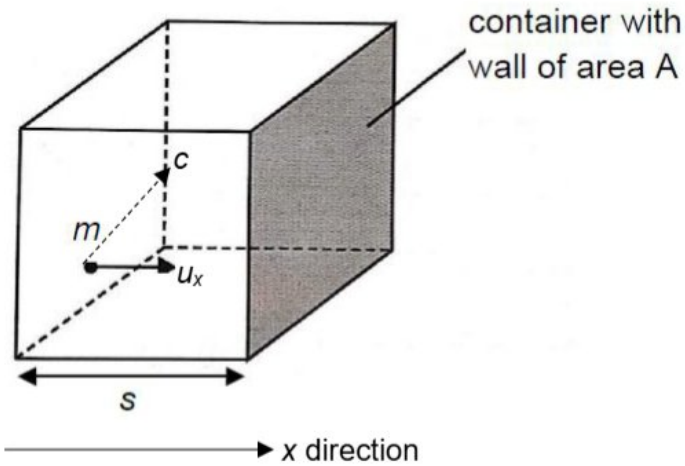


Fig. 4.1

Derive, in terms of m, u_x and s,

1. the change in momentum of the molecule when it collides with the wall on the right, [1]

2. the time between consecutive collisions with the wall on the right. [1]

(ii) Hence show that the average force F exerted by the molecule on the wall on the right is given by $F = \frac{mu_x^2}{s}$. [1]

(iii) Each molecule of the gas has its own velocity c and velocity component u_x in the x direction.

1. State the equation that relates the mean square speed of the molecules, $<c^2>$, to the mean square speed due their velocity components in the x direction, $<u_x^2>$. [1]

2. Using your answers to **(a)(ii) and (a)(iii)1**, derive the expression $pV = \frac{1}{3}Nm\langle c^2 \rangle$, where p is the pressure on the wall, V is the volume of the container, N is the number of molecules and $\langle c^2 \rangle$ is the mean square speed of the molecules. [2]

(b) A cylinder, shown in Fig. 4.2, fitted with a piston contains ideal gas at a constant temperature of 20 °C. When the volume V is 0.50×10^{-3} m^3, the pressure p in the cylinder is 20×10^4 Pa.

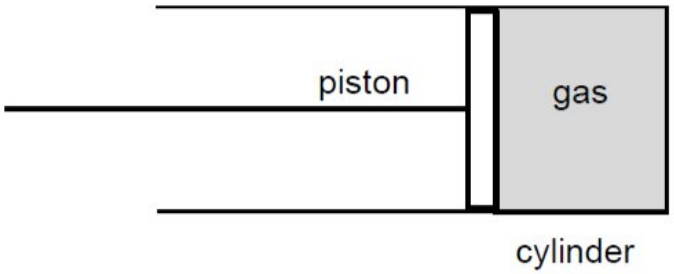


Fig. 4.2

An experiment was conducted and the data was plotted in Fig. 4.3

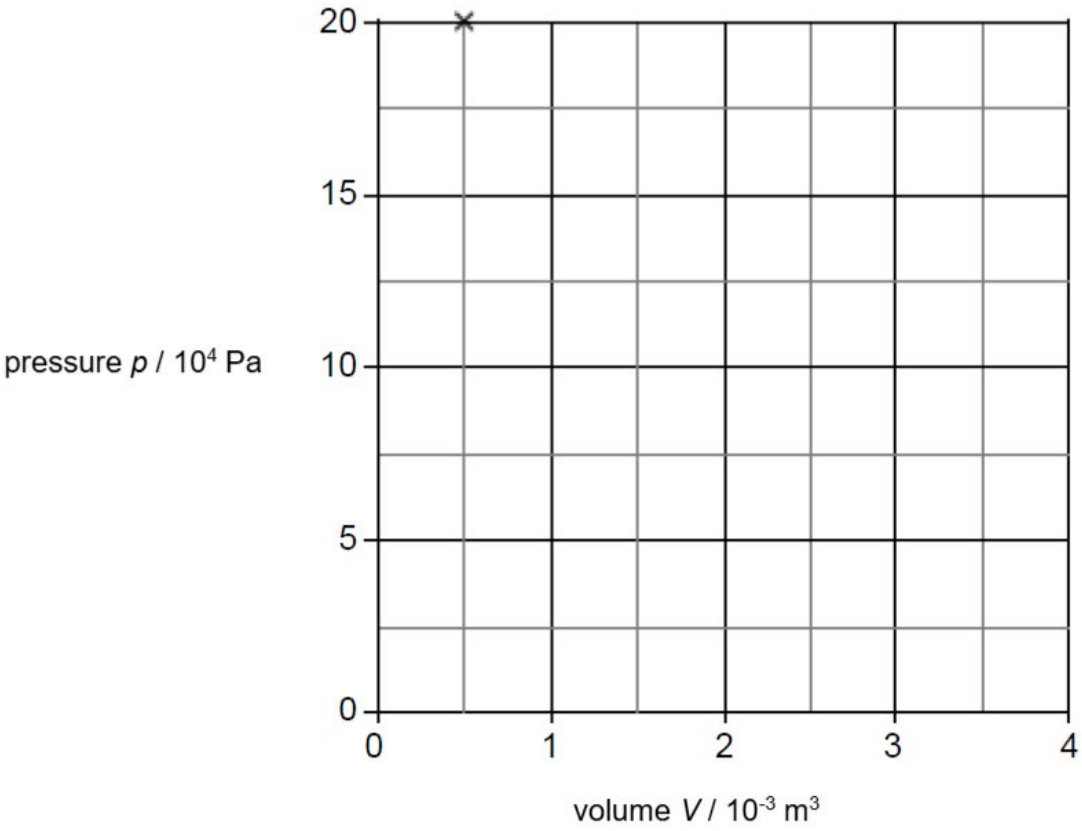


Fig. 4.3

(i) By plotting two additional points, draw a graph on Fig. 4.3 to show the relationship between p and V as the piston is slowly pulled out such that the temperature of the gas remains constant. [1]

(ii) Calculate the number of gas molecules in the cylinder.

number of gas molecules = ... [2]

(iii) Determine the total kinetic energy of the gas molecules.

total kinetic energy = ... J [1]

[Total: 10]